Identification theory

Lecture 1. From an engineering question to an experiment

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What this course is about

Identification theory studies how to determine unknown parameters in mathematical models from observations.

The course is for fourth-year Automation bachelor students, in semester 7. It has **40 hours: 16 hours of lectures and 24 hours of labs and consultations**.

We study both **offline identification**, using an already collected record, and **online identification**, updating estimates as new data arrive.

In the lab sessions, we work with **MATLAB and Simulink**. Lecture presentations and lab guides are the main course materials.

What you should be able to do

By the end of the course, you should understand the basics of system identification and selected offline and online methods.

You should be able to **estimate parameters of linear regression models** and transform linear dynamic models into regression form for parameter estimation.

You should also be able to transform **some nonlinear problems** into linear regression. This does not mean that every nonlinear problem has such a transformation.

You should be able to **compare the quality of identified models**. A parameter estimate is useful when the resulting model serves its intended task.

Two parts: offline and online identification

First part: offline identification. We begin with general concepts, approximation and regression, then study the least-squares estimation method and its computational use.

Second part: online identification. We study gradient descent, dynamic regressor extension and mixing (DREM), and least-squares methods for online identification.

Course practice and today’s starting point

Practice connects the two parts. Lab topics include approximation, parameter estimation, dynamic models, excitation, estimator comparison and predictive models.

Today is the starting point: formulate a modelling task, choose an experiment, determine parameters in an ideal example and check the model. The later methods are course topics, not prerequisites for this lecture.

Coursework, assessment and communication

The syllabus assigns **80 points to laboratory work** and **20 points to the final test**, for a total of 100 points.

Lab 1: 20 points. Lab 2: 30 points. Lab 3: 30 points. Final test: 20 points. The pass-score range is 60–100 points; the complete checkpoint requirements are specified in the syllabus.

At least **70% attendance** is required. If you expect repeated absence or lateness, contact the student office as described in the course policies.

Questions can be sent by **email or the course DingTalk group**. Instructors answer on weekdays, two or three times a week, between **17:00 and 19:00 Moscow time**.

Today: from an engineering question to an experiment

M01–M03 The purpose, the task and the measured quantities

M04–M09 Model structure, dynamics and the RL assumptions

M10–M15 An ideal experiment: determine R and L

M16–M20 Design and record an informative experiment

M21–M23 Check the model on new data

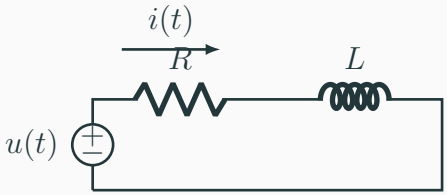
M24 Next: many imperfect measurements and least squares

Reference appendices A01–A12 and further explanations are for self-study.

An equation is not yet a predictive model

Identification theory · Lecture 1: From an engineering question to an experiment

Consider an electrical circuit containing a resistor and an inductor. A physical law gives us an equation, but the resistance and inductance may still be unknown.



An equation is not yet a predictive model

We want to use measured voltage and current to determine these parameters. We then want to calculate the current for a new voltage signal.

Today we will complete one example: define the task, choose a model, design an experiment, determine its parameters and check its response on new data.

Learning goal: Explain not only how parameters are calculated, but also why the chosen experiment contains the information needed to calculate them.

Identify a model for a stated purpose

System identification uses observations of a system to determine a mathematical model of its behaviour.

A model is useful only in relation to a task. For our circuit, compare two tasks:

Task A — final current. Calculate the current after a constant voltage has been applied for a long time.

Task B — changing current. Calculate how the current changes after the voltage changes.

Task A may need only the final input–output relation. Task B also needs the speed of the response.

Identify a model for a stated purpose

For this lecture, our target is Task B: reproduce the current during voltage changes within the selected operating range. We must check both the final level and the transition towards it.

Key point: State what the model must reproduce before deciding whether it is good enough.

Separate the input, the output, measurements and parameters

The **input** is the voltage applied to the circuit: $u(t)$, measured in volts (V). The **output** is the circuit current: $i(t)$, measured in amperes (A).

The unknown **parameters** are the resistance R, in ohms (Ω), and the inductance L, in henries (H). We assume that they remain constant during an experiment.

A sensor gives samples at times t_k. Write the measured current as

$$y_k = i(t_k) + \delta_k.$$

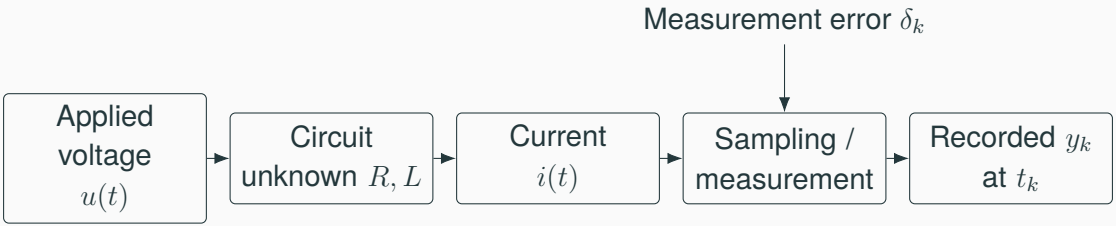
Separate the input, the output, measurements and parameters

Here y_k is the sensor reading and δ_k is measurement error. We do not assign a probability distribution to this error in this lecture.

The record must connect each voltage sample, current sample and time stamp. An unknown parameter is not the same thing as an unmeasured output.

Key point: Distinguish the physical quantity, its recorded value and the parameter to be determined.

Separate the input, the output, measurements and parameters



$$y_k = i(t_k) + \delta_k.$$

The measurement error enters the observation stage, not the circuit input. Unknown R, L are parameters; $i(t)$ is a physical response and y_k is a recorded value.

Choosing a structure is different from choosing its parameters

A **model structure** specifies the form of the relations and the quantities that appear in them. A **parameter value** selects a particular model within that structure.

Two candidate descriptions of our circuit are

$$\text{equilibrium relation:} \quad Ri = u,$$

$$\text{dynamic model:} \quad L\frac{di}{dt} + Ri = u.$$

Choosing a structure is different from choosing its parameters

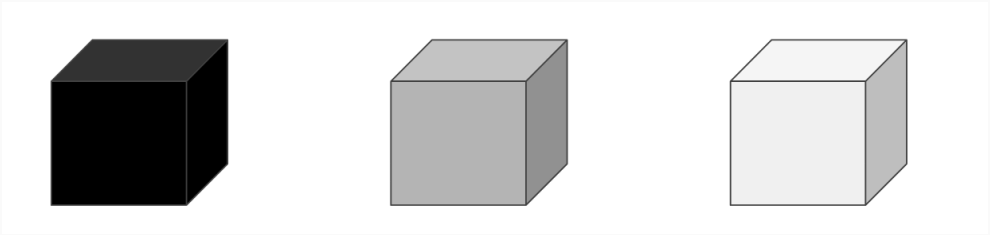
The first relation has no term describing how current changes. The second includes the current derivative di/dt, meaning its rate of change.

Choosing numerical values for R cannot add the missing dynamic term to the first relation. Choosing the second structure still leaves the numerical values of R and L to be determined.

Key point: A parameter calculation cannot automatically repair an unsuitable model structure.

Physical knowledge and data play different roles

A **white-box approach** builds a model mainly from physical laws and independently available physical information.



Black-box Grey-box White-box

Physical knowledge and data play different roles

A **grey-box approach** uses physical knowledge to select a structure and measurements to determine unknown parameters. Our RL example follows this approach: the equation is given by physics; R and L are determined from the response.

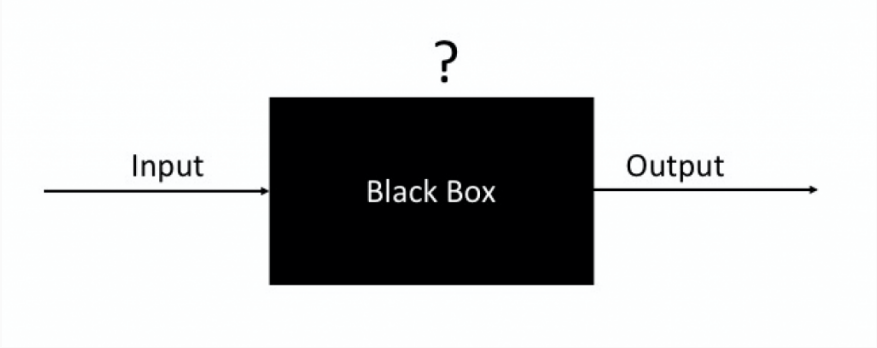
A **black-box approach** selects a mathematical input–output description and fits it to data without explicitly deriving all its terms from the internal physics.

Physical knowledge and data play different roles

These labels describe how information is used. They do not, by themselves, determine computational cost, accuracy or flexibility. A black-box model does not have to be a neural network.

Key point: Physical equations and experimental measurements are complementary sources of information.

Physical knowledge and data play different roles



The input and output are visible; the internal mechanism is not specified.

A dynamic model has memory

A **static model** gives the output from the current input without using past values or an internal state. For example, $\hat{i}(t) = u(t)/R$ is a static description.

A **dynamic model** includes memory. The RL current depends on the voltage history and on the initial current, not only on the voltage at the present instant.

For example, two identical RL circuits can have the same voltage now but different currents because their earlier voltages were different.

A dynamic model has memory

A dynamic system can also be at equilibrium. When its current becomes constant, the derivative is zero, but the system has not lost its dynamic properties.

Key point: “Dynamic” means that history matters; it does not mean that the output must always be changing.

Model time, sampling and updating are different choices

A **continuous-time model** describes a response for time t. A **discrete-time model** relates values indexed by samples, such as k and $k+1$.

Sampling means recording measurements at selected times. With uniform sampling,

$$t_k = k\Delta t,$$

where Δt is the sampling interval in seconds.

Offline identification processes an already collected record. **Online identification** updates a model as new measurements become available.

Model time, sampling and updating are different choices

These are different questions. In our example, the physical model is continuous-time, the measurements are sampled, and the parameter calculation is offline.

A continuous-time model does not require online updating. Sampled measurements do not imply that the physical current changes in jumps.

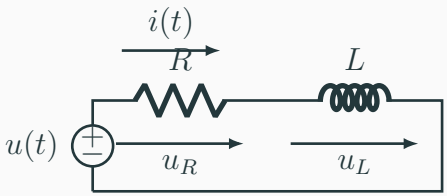
Key point: Keep the model representation, the measurement record and the updating procedure separate.

State the assumptions before using the RL equation

We use the model

$$L\frac{di(t)}{dt} + Ri(t) = u(t), \qquad R > 0, \quad L > 0.$$

For the worked example, assume a series resistor and ideal inductor, constant parameter values, a known applied voltage and an initial current $i(0) = 0$.



Kirchhoff's voltage law: $u(t) = u_R + u_L$, with $u_R = Ri(t)$ and $u_L = L\,di(t)/dt$.

State the assumptions before using the RL equation

We neglect additional circuit effects in this teaching model. We must revisit these assumptions if the model misses an important part of the response.

With fixed R, L and zero initial current, this model is linear in its response to the input: scaling the voltage scales the current. This does not mean that the response formula is linear in the unknown parameters.

Key point: A physical equation is used together with assumptions, not instead of them.

Identification is an iterative process

1. **Define the purpose.** Specify what the model must calculate and under which conditions.
2. **Select a model structure and an estimation approach.** State the unknown quantities and the assumptions.
3. **Design and record an experiment.** Choose an input and measurements that reveal the required behaviour.
4. **Determine the parameters.** Use the recorded information and the selected method.

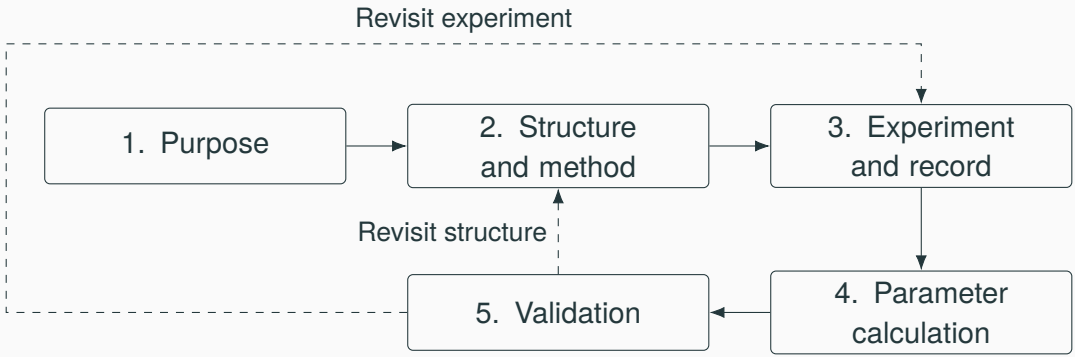
Identification is an iterative process

5. Validate the model. Compare its response with a new record and judge it against the original purpose.

If validation fails, revisit the data, the experiment, the parameter calculation or the model structure. The appropriate revision depends on the evidence.

Key point: A numerical parameter vector is an intermediate result, not the end of identification.

Identification is an iterative process



Validation can lead back to the model structure or the experiment. The return path is selected from the evidence; a parameter vector alone does not complete identification.

Equilibrium data determine resistance, but not inductance

Apply a constant voltage U and consider the final equilibrium current i_∞.

Step 1. At equilibrium the current no longer changes, so $di/dt = 0$.

Step 2. Substitute this condition into the model:

$$L \cdot 0 + R i_\infty = U.$$

Step 3. Rearrange the remaining relation:

$$R = \frac{U}{i_\infty}.$$

Equilibrium data determine resistance, but not inductance

The inductance L does not appear. Different positive inductances are compatible with the same equilibrium voltage and current.

Even several equilibrium voltage–current pairs cannot determine L within this model. They can provide information about the static relation, but not about the missing transient behaviour.

Key point: More data of the same kind need not reveal every unknown parameter.

Different inductances can give the same final current

Use the same resistance $R = 2\,\Omega$ and constant voltage $U = 4\,\mathrm{V}$, but compare three inductances:

$$L = 0.2\,\mathrm{H}, \quad 0.5\,\mathrm{H}, \quad 1.0\,\mathrm{H}.$$

All three responses start from zero current. All approach the same final level:

$$i_\infty = \frac{4}{2} = 2\,\mathrm{A}.$$

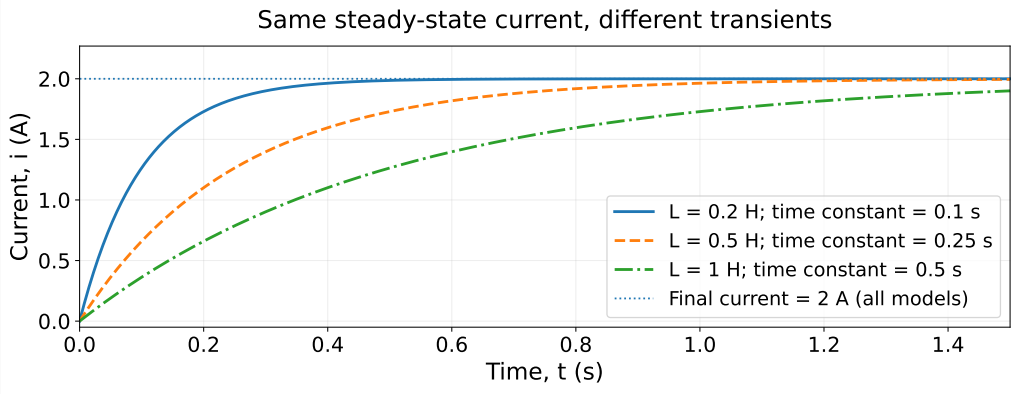
The curves differ during the transition. A larger inductance produces a slower change when the resistance is fixed.

Different inductances can give the same final current

Therefore, record the current during the transition, not only after it has almost settled. The shape of the transient distinguishes models that have identical equilibrium behaviour.

Key point: Choose observations where the candidate models make different predictions.

Different inductances can give the same final current



Synthetic ideal responses: R = 2 ohm, U = 4 V, i(0) = 0. All curves approach 2 A; L = 0.2, 0.5 and 1.0 H give different transients.

The step response has a final level and a time scale

For a voltage step from zero to U at $t = 0$, with $i(0) = 0$, the model gives

$$i(t) = \frac{U}{R}\left(1 - e^{-t/\tau}\right), \qquad \tau = \frac{L}{R}, \qquad t \geq 0.$$

The **final level** is $i_\infty = U/R$. It describes how large the current eventually becomes.

The **time constant** τ, measured in seconds, describes the speed of the transition. The fraction t/τ has no units.

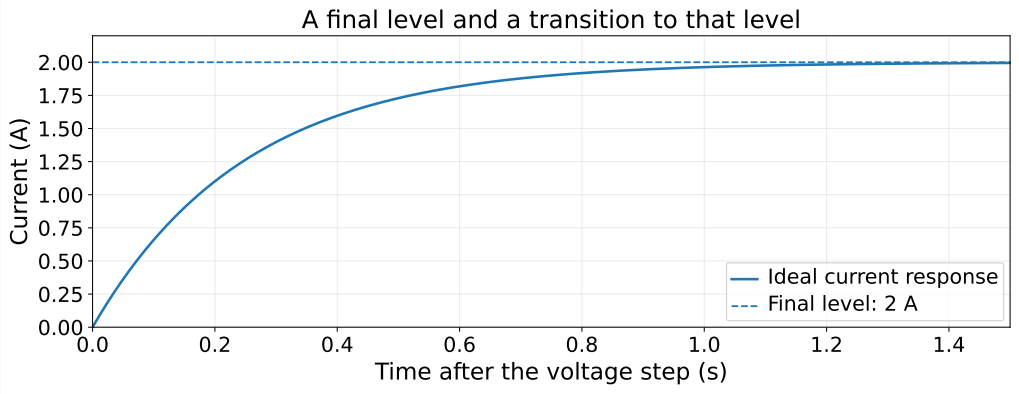
The step response has a final level and a time scale

We can therefore use two different response features: the final level to determine R, and the time scale to determine L/R.

The derivation is available in Appendix A03; we now use the formula to interpret the experiment.

Key point: Different response features provide information about different parameter combinations.

The step response has a final level and a time scale



Synthetic ideal response to a 4 V step, with zero initial current. The final level and transition are shown before the parameter calculation.

Step 1: recover the resistance from the final level

Suppose the ideal response to $U = 4\,\mathrm{V}$ has final current $i_\infty = 2\,\mathrm{A}$.

The equilibrium relation is

$$Ri_\infty = U.$$

Divide by the final current:

$$R = \frac{U}{i_\infty} = \frac{4\,\mathrm{V}}{2\,\mathrm{A}} = 2\,\Omega.$$

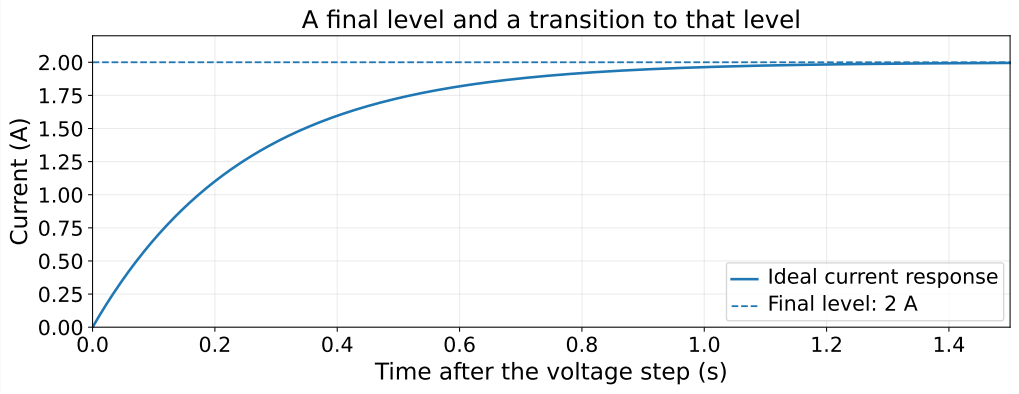
This calculation uses the final level, not the speed of the curve. We have recovered one parameter; the inductance remains unknown.

Step 1: recover the resistance from the final level

In an actual finite record, the plateau must be estimated from measurements. The last sample alone is not automatically the true equilibrium value.

Key point: Match the measured feature to the relation that contains the parameter.

Step 1: recover the resistance from the final level



Synthetic ideal step response: the asymptotic level is 2 A. With U = 4 V, the level gives R = 2 ohm; a finite last sample is not the asymptote.

Step 2: read the time constant at 63.2% of the change

Substitute $t = \tau$ into the zero-initial-current response:

$$i(\tau) = i_\infty(1 - e^{-1}) \approx 0.632\, i_\infty.$$

For $i_\infty = 2\,\mathrm{A}$, the target current is

$$i(\tau) \approx 0.632 \times 2 = 1.264\,\mathrm{A}.$$

Find this current on the vertical axis, move horizontally to the response curve, and then move down to the time axis.

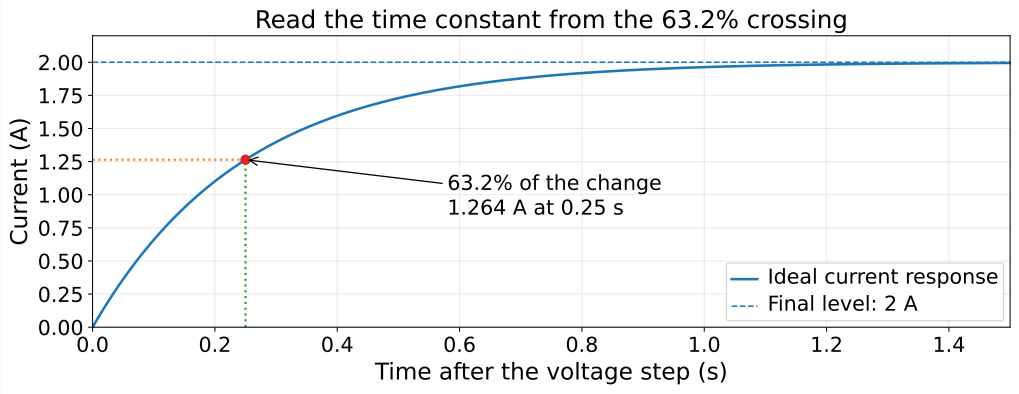
Step 2: read the time constant at 63.2% of the change

For the ideal curve shown, the crossing occurs at $t = 0.25\,\mathrm{s}$. Therefore, $\tau = 0.25\,\mathrm{s}$.

The 63.2% refers to the change from the initial value to the final value. It equals 63.2% of the final current here only because the initial current is zero.

Key point: A time constant is a measurable feature of this assumed response shape.

Step 2: read the time constant at 63.2% of the change



Synthetic ideal zero-start response. The horizontal and vertical guides meet at 1.264 A and 0.25 s. The target is 63.2% of the change, not a universal current level.

Step 3: recover the inductance and check the result

We have obtained $R = 2\,\Omega$ and $\tau = 0.25\,\mathrm{s}$. Since $\tau = L/R$,

$$L = R\tau = 2\,\Omega \times 0.25\,\mathrm{s} = 0.50\,\mathrm{H}.$$

The identified ideal model is

$$0.50\frac{d\hat{i}}{dt} + 2\hat{i} = u, \qquad \hat{i}(0) = 0.$$

Step 3: recover the inductance and check the result

A hat marks the model-calculated current. For the original 4 V step,

$$\hat{i}(t) = 2\left(1 - e^{-t/0.25}\right)\ \mathrm{A}.$$

Check the features: $\hat{i}(0) = 0$, $\hat{i}(0.25) \approx 1.264\,\mathrm{A}$, and $\hat{i}(t) \to 2\,\mathrm{A}$.

This checks consistency with the information used in the calculation. It is not yet a test on new data.

Key point: Recover the parameters, then distinguish consistency checks from independent validation.

Choose an input that exposes the behaviour you need

In an **active experiment**, we deliberately choose the input to obtain useful information. In a **passive experiment**, we use observations from normal operation without imposing a special identification input.

For our RL task, equilibrium values reveal resistance, while a recorded transition also reveals the time constant. A new sequence of voltage levels can test the calculated response after several changes.

Choose an input that produces observable differences between plausible models and stays within the allowed operating conditions.

Choose an input that exposes the behaviour you need

A step is sufficient for the stated ideal RL example, but it is not a universal input for every system. Existing operating records may also be useful when they contain the required changes and reliable measurements.

Key point: Input design depends on the model, the unknown quantities and the intended use.

Sampling changes how well the transient is represented

The model current varies continuously, but the record contains selected samples. Compare the same ideal response with $\tau = 0.25\,\mathrm{s}$:

Coarse record: $\Delta t = 0.50\,\mathrm{s}$. The first sample after the change is already two time constants later.

Finer record: $\Delta t = 0.025\,\mathrm{s}$. Ten sampling intervals fit within one time constant.

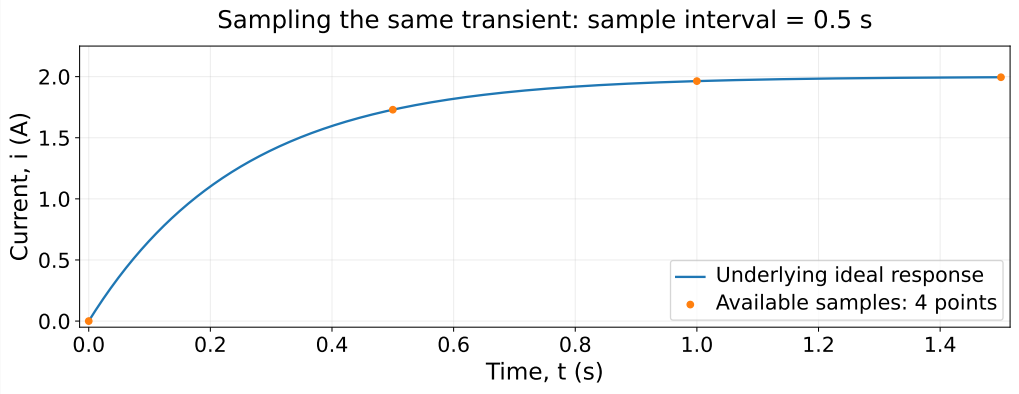
The finer record shows the shape of the transition more directly. The coarse record provides far fewer points near the 63.2% crossing.

Sampling changes how well the transient is represented

These numbers illustrate this model; they are not universal sampling requirements. Sampling, input timing, sensor behaviour and measurement error must be considered together.

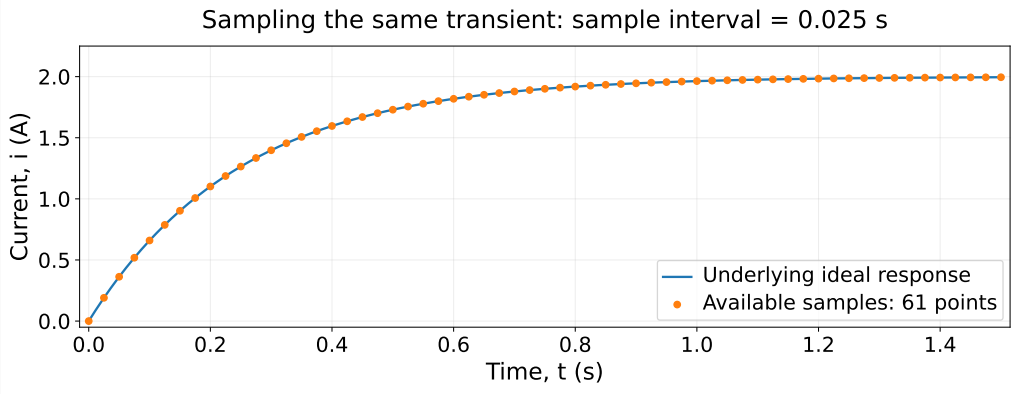
Key point: Record the time scale you need to identify, rather than only the final level.

Sampling changes how well the transient is represented



Synthetic ideal response: time constant 0.25 s; sampling interval 0.50 s. The first sample after the step is two time constants later.

Sampling changes how well the transient is represented



Same synthetic response and the same axes as the coarse record; interval 0.025 s. Ten sampling intervals fit within one time constant; this is not a universal requirement.

Choose the recording duration and the initial condition

The initial current affects the whole transient. Our numerical calculation assumes $i(0) = 0$; another initial value requires a different response formula.

For the zero-initial-current RL response, the remaining fraction of the change at time T is

$$e^{-T/\tau}.$$

At $T = 5\tau$, the response has completed

$$1 - e^{-5} \approx 99.3\%$$

Choose the recording duration and the initial condition

of the total change. With $\tau = 0.25\,\mathrm{s}$, this is $T = 1.25\,\mathrm{s}$.

A finite record therefore approaches, but does not exactly reach, the ideal final value. If the time scale is initially unknown, use prior knowledge or a preliminary experiment to choose the duration and sampling.

Key point: Plan what happens before, during and after the informative transition.

Record what actually happened in the experiment

A useful record contains more than the output curve.

Keep the applied voltage, measured current and matching time stamps. Record the units, initial condition, switching times, operating conditions and the source of each signal.

Distinguish a voltage **command** from the voltage actually applied to the circuit. They need not be identical in an imperfect setup.

Check sensor offset, range limits, missing samples and the time alignment between input and output. Record any filtering or other processing that changes the signals.

Record what actually happened in the experiment

Also specify how the input behaves between samples. In our simulation, each voltage level is held until the next switch.

Key point: Unrecorded details can later look like errors in the model parameters.

Write a concrete specification before collecting data

Purpose: Reproduce the current during voltage changes using the assumed RL model.

Unknown quantities: Constant positive resistance R and inductance L.

Teaching identification record: A voltage step from 0 to 4 V, zero initial current, sampling interval 0.025 s, and 1.5 s of response data.

Recorded information: Applied voltage, current readings, time stamps, units, initial condition, switching convention and processing history.

Validation plan: A separate sequence of voltage levels inside 0–4 V. Freeze the identified parameters before examining this response.

Write a concrete specification before collecting data

Scope: These are chosen simulation settings, not a validated protocol for physical hardware. A real experiment additionally needs apparatus-specific limits and an acceptance criterion linked to the application.

Key point: Make experimental decisions explicit and reproducible.

Validate using a new record and fixed parameters

Keep $\hat{R} = 2\,\Omega$ and $\hat{L} = 0.50\,\mathrm{H}$ fixed. Apply a different voltage sequence, starting from zero current:

$$0 \longrightarrow 2 \longrightarrow 4 \longrightarrow 1 \longrightarrow 0 \quad \mathrm{V}.$$

The switches occur at 0.25, 0.75, 1.50 and 2.00 s. Record the response until 3.00 s with interval 0.025 s.

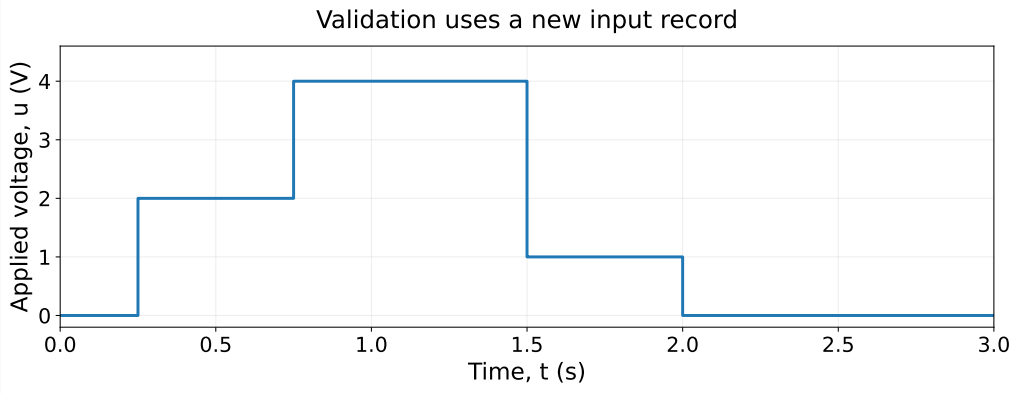
Calculate the model response to this input without refitting its parameters. Compare it with the new current readings.

Validate using a new record and fixed parameters

Here the readings are synthetic: we add simulated measurement error to the ideal reference response. The parameters came from the earlier ideal step example, not from fitting this new record.

Key point: A validation record checks a frozen model; it is not additional fitting data for the same check.

Validate using a new record and fixed parameters



Synthetic validation input, held constant between switches at 0.25, 0.75, 1.50 and 2.00 s. Initial current is zero; record ends at 3.00 s. Model parameters remain fixed.

Compare the predictions during transitions and plateaus

Compare two models using the same applied voltage:

$$\text{static:} \quad \hat{i}_{\text{static}}(t) = \frac{u(t)}{\hat{R}},$$

$$\text{dynamic:} \quad \hat{L}\frac{d\hat{i}_{\text{dynamic}}}{dt} + \hat{R}\hat{i}_{\text{dynamic}} = u.$$

The static model changes its predicted current immediately when the voltage changes. The dynamic model carries its current state through each transition.

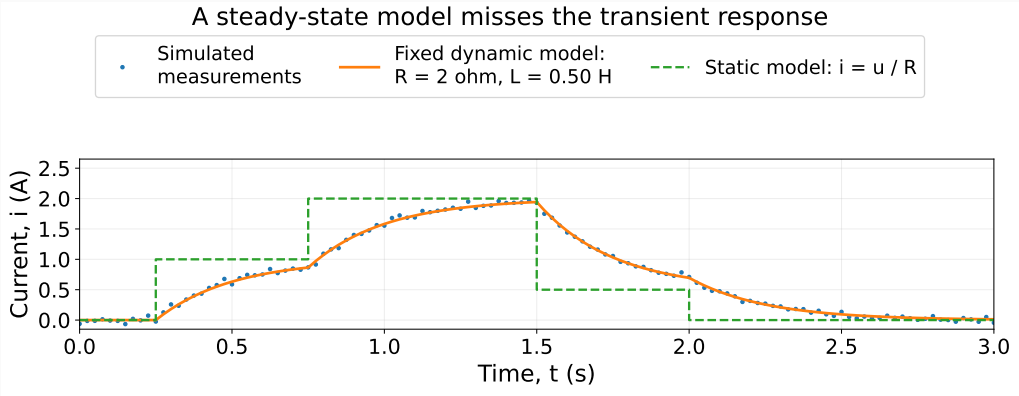
Compare the predictions during transitions and plateaus

On the synthetic record, the dynamic model follows the transition shape. The static model describes equilibrium levels but misses the transitions towards them.

For our stated task, transition behaviour matters. Therefore, equilibrium agreement alone is insufficient.

Key point: Evaluate a model where the intended task needs it to be accurate.

Compare the predictions during transitions and plateaus



Synthetic current readings with added measurement error. Fixed dynamic model: R = 2 ohm, L = 0.50 H. Static prediction: u/R. The current state continues through each switch.

Use the mismatch to decide what to examine next

Define the **simulation residual** at sample k:

$$e_k = y_k - \hat{i}(t_k).$$

It is the measured current minus the current calculated by the fixed model using the recorded input and initial condition.

In our comparison, the static model leaves systematic errors near voltage changes. It is missing behaviour important for the task.

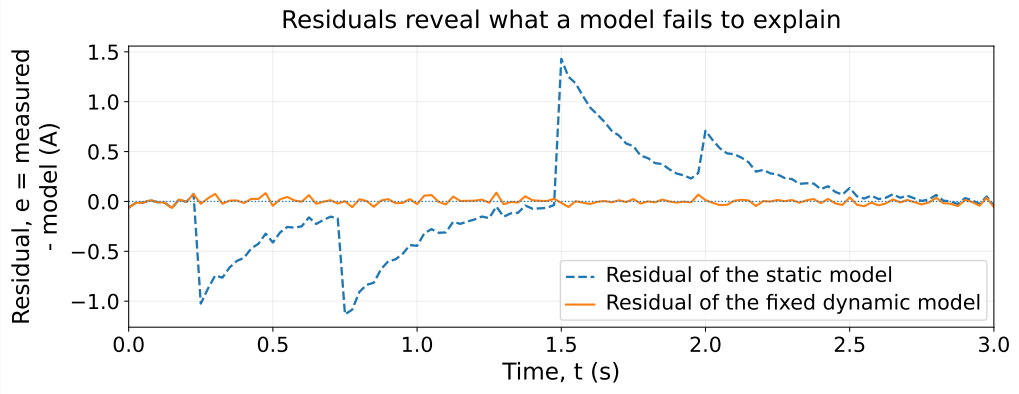
Use the mismatch to decide what to examine next

In another experiment, a mismatch could also come from a sensor offset, wrong time alignment, a wrong initial condition, parameter error or an unsuitable structure.

A residual pattern is evidence to investigate; it does not uniquely identify its cause. We are not applying a statistical white-noise test here.

Key point: Validation guides the next model or experiment decision; it does not automatically explain every error.

Use the mismatch to decide what to examine next



Synthetic simulation residuals: measured minus model-calculated current, in amperes. Fixed model parameters; no one-step prediction error or statistical whiteness test is plotted.

From an ideal curve to many imperfect measurements

Our example connected five decisions: define the purpose, choose a structure, obtain informative data, determine parameters and validate the result.

For the ideal RL response, the final current and the time constant gave us R and L. Equilibrium data alone could not provide both.

Real measurements do not lie exactly on an ideal curve. Reading one final level or one crossing can give uncertain results, and different observations may suggest different parameter values.

We need a method that combines many observations and states what “best agreement” means.

From an ideal curve to many imperfect measurements

Next lecture: Least squares as a deterministic approximation problem. We will introduce the criterion and calculate parameters from a given data set. Probability assumptions and statistical properties come later.

Key point: Good parameter calculation begins with a clear model and useful data.

Symbol guide: quantities in the experiment

t, t_k, k. Continuous time, the time of sample k, and the sample index. Time is measured in seconds.

$u(t)$, U. Applied voltage waveform and the constant voltage after a step. Unit: volt (V).

$i(t)$, i_0, i_∞. Current, initial current and limiting equilibrium current. Unit: ampere (A).

Symbol guide: quantities in the experiment

R, L, $\tau = L/R$. Resistance, inductance and time constant. Units: ohm (Ω), henry (H) and second (s).

Δt, T. Sampling interval and recording duration. Unit: second (s).

An overdot means differentiation with respect to time: $\dot{i} = di/dt$. The ratio t/τ is dimensionless, so it can appear inside an exponential.

Word guide: observations, models and checking

Input / output: An applied or observed cause signal and the response quantity selected for the task.

Measurement: A recorded observation; it may differ from the corresponding ideal value.

Parameter: A quantity that selects one model within a specified structure. Here R and L are constant during each experiment.

Transient: The changing part of a response following a change in conditions.

Word guide: observations, models and checking

Equilibrium / steady state: A regime in which this circuit's current is constant under constant voltage. "Steady state" can have a broader meaning in other problems.

Estimate / model output: A value determined from data / a value calculated by the model. A hat marks a calculated or estimated quantity.

Validation: Checking a selected model against observations and the stated purpose.

Derive the ideal step response

For constant voltage U and positive constant R and L, start from

$$L\dot{i} + Ri = U.$$

1. **Find the equilibrium value:** $i_\infty = U/R$.
2. **Subtract it:** define $z = i - i_\infty$. Because i_∞ is constant,

$$\dot{z} = -\frac{R}{L}z.$$

Derive the ideal step response

3. Solve the homogeneous equation:

$$z(t) = z(0)e^{-Rt/L}.$$

4. Apply $i(0) = 0$: $z(0) = -i_\infty$. Therefore,

$$i(t) = \frac{U}{R}\left(1 - e^{-Rt/L}\right) = i_\infty\left(1 - e^{-t/\tau}\right), \quad \tau = L/R.$$

The exponential decays because R/L is positive.

A nonzero initial current changes the 63.2% target

For a constant input U and initial current $i(0) = i_0$, the response is

$$i(t) = i_\infty + (i_0 - i_\infty)e^{-t/\tau}.$$

At one time constant,

$$i(\tau) = i_0 + (1 - e^{-1})(i_\infty - i_0).$$

A nonzero initial current changes the 63.2% target

Example: Let $i_0 = 0.5\,\mathrm{A}$, $i_\infty = 2\,\mathrm{A}$. The total change is $1.5\,\mathrm{A}$, so

$$i(\tau) = 0.5 + 0.63212056 \times 1.5 \approx 1.448\,\mathrm{A}.$$

Reading $0.632 \times 2 = 1.264\,\mathrm{A}$ would use the wrong target for this initial condition.

The same expression also describes a decreasing response when $i_0 > i_\infty$.

Two meanings of “linear” must not be confused

Linear in the input: A zero-initial-state input–output map obeys superposition. For a fixed RL model, doubling the voltage doubles the current response.

Affine in the input: The relation $y = au + b$ with $b \neq 0$ has a nonzero offset and is not a linear map of u in the strict superposition sense.

Linear in the parameters: With observed u treated as fixed, $y = au + b$ is linear in the unknown coefficients a and b.

Two meanings of “linear” must not be confused

Similarly, $y = \theta_0 + \theta_1 u + \theta_2 u^2$ is linear in its coefficients even though its dependence on u can be nonlinear.

The term “linear regression” in the next lecture concerns how unknown coefficients enter the model. It does not require a straight-line dependence on every original input variable.

Sampling and quantisation describe different operations

Sampling selects observation times. A sampled record can be written as $y_0, y_1, \ldots$ at times $t_0, t_1, \ldots$.

Quantisation represents measured amplitudes using a finite set of levels. It concerns resolution in the recorded values, not the time interval between them.

For example, changing the sampling interval from 0.50 s to 0.025 s adds observations during a transition. It does not, by itself, improve the amplitude resolution of a sensor.

Sampling and quantisation describe different operations

A mathematical discrete-time model is a relation between indexed values. Its amplitudes do not have to be restricted to a finite set in the mathematical description.

Keep these questions separate from whether parameters are updated offline or online.

How the piecewise-constant validation response is computed

Suppose $u(t) = u_k$ is held constant on $[t_k, t_{k+1})$. Define

$$\alpha_k = e^{-R(t_{k+1}-t_k)/L}.$$

The exact ideal RL update is

$$i_{k+1} = \alpha_k i_k + (1 - \alpha_k)\frac{u_k}{R}.$$

This expression advances the current from one sample time to the next while carrying its previous value forward.

How the piecewise-constant validation response is computed

For $\Delta t = 0.025\,\mathrm{s}$, $R = 2\,\Omega$ and $L = 0.5\,\mathrm{H}$,

$$\alpha = e^{-0.1} \approx 0.904837.$$

At a voltage switch the current remains continuous. The newly selected u_k affects the following interval, not an instantaneous reset of i_k.

A sensor offset can change the inferred resistance

Suppose the true equilibrium current is $2.0\,\mathrm{A}$, but an unknown sensor offset adds $0.1\,\mathrm{A}$. The recorded equilibrium value is then $2.1\,\mathrm{A}$.

If we ignore the offset and apply the ratio formula,

$$\hat{R} = \frac{4}{2.1} \approx 1.905\,\Omega,$$

instead of the true $2.0\,\Omega$ used in this synthetic example.

A sensor offset can change the inferred resistance

The relative error is approximately

$$\frac{1.905 - 2}{2} \times 100\% \approx -4.76\%.$$

More readings of the same biased level do not automatically remove this unknown offset.

Lesson: An accurate calculation can still produce a wrong parameter when the measurement assumptions are wrong.

Exact equilibrium is different from a late finite sample

In the ideal zero-start response,

$$i(T) = \frac{U}{R}\left(1 - e^{-RT/L}\right).$$

At every finite positive T, this value still depends on L. Its sensitivity is

$$\frac{\partial i(T)}{\partial L} = -\frac{UT}{L^2}e^{-RT/L}.$$

As T becomes large, this sensitivity approaches zero.

Exact equilibrium is different from a late finite sample

At exact equilibrium the relation is $i_\infty = U/R$, which contains no L at all.

Therefore distinguish two statements: equilibrium-only data do not determine L in this model; late transient data may contain only a small observable effect of L when measurement errors are present.

The main lecture uses the first statement to explain why recording the transition matters.

Worked practice: identify another ideal RL response

Given: A zero-start response to $U = 3\,\mathrm{V}$, final current $i_\infty = 1.5\,\mathrm{A}$, and a 63.2% crossing at $0.20\,\mathrm{s}$.

1. Resistance:

$$R = U/i_\infty = 3/1.5 = 2\,\Omega.$$

2. Time constant: The crossing is measured from the input change, so $\tau = 0.20\,\mathrm{s}$.

Worked practice: identify another ideal RL response

3. **Inductance:**

$$L = R\tau = 2 \times 0.20 = 0.40\,\mathrm{H}.$$

4. **Check the target current:** $i(\tau) = 0.63212056 \times 1.5 \approx 0.948\,\mathrm{A}.$

5. **Write the response:** $i(t) = 1.5(1 - e^{-t/0.20})\,\mathrm{A}.$

The given final current alone would not determine L.

Common incorrect conclusions, with explanations

“More samples always identify every parameter.” No. Exact equilibrium removes L from the RL input–output relation.

“A constant input is always useless.” No. The transition after a step contains information about this model’s time constant.

“A dynamic model must always have a changing output.” No. A dynamic system can be at equilibrium.

“A continuous-time model requires online identification.” No. We used a continuous-time equation with sampled data processed offline.

Common incorrect conclusions, with explanations

“A good fit proves the model is correct everywhere.” No. Checking one range and one class of inputs does not establish universal validity.

“Least squares requires Gaussian noise.” No. The deterministic minimisation problem in the next lecture can be defined without a noise distribution. Statistical interpretations require their own assumptions.

Reading guide: the next useful explanation

For the circuit equation and response: OpenStax, *University Physics, Volume 2*, Section 14.4, “RL Circuits”. Read the differential equation, step response and definition of the time constant. [T2]

For the identification process: MathWorks, “System Identification Overview”. Read the introduction, grey-box modelling and how to evaluate a model. Software commands are not required for this lecture. [T1]

For checking a model on new data: MathWorks, “Validating Models After Estimation”. Focus on the separation of estimation and validation data. [T5]

Reading guide: the next useful explanation

For the next lecture: Boyd and Vandenberghe, *Introduction to Applied Linear Algebra*, Section 2.1 and Chapter 13. [T7]

The numerical RL examples in these slides are newly constructed teaching illustrations [D1], not experimental results taken from these sources.

D1 — Reference and reading scope

New synthetic RL teaching example and author-derived calculations

Original numerical illustration prepared for this lecture, not an external experimental source.

R=2 ohm, L=0.5 H, U=4 V, tau=0.25 s. All CSV data and graphs are generated locally from the stated equations. Ideal feature recovery is followed by a synthetic noisy validation record. No noisy-data parameter fit is performed.

S1 — Reference and reading scope

Identification Theory — course syllabus, Automation AT4, 2026

User-provided course syllabus; source for the course orientation only.

Page 1: course structure, tools and learning objectives. Pages 2–3: communication, attendance and assessment. Pages 4–5: course topics. Read the course description, learning objectives and schedule for the overall course. The supplied syllabus is included with the lecture materials.

T1 — Reference and reading scope

MathWorks — System Identification Overview

Technical reference for the identification task, grey-box structure and validation purpose.

Read the introductory definition, grey-box discussion and How Do You Know If Your Model Is Good Enough? It also distinguishes continuous/discrete representations and input behaviour between samples. The source does not prescribe our 24-unit sequence or numerical experiment.

Open the source directly

T2 — Reference and reading scope

Ling, Moebs and Sanny — University Physics, Volume 2, Section 14.4: RL Circuits (OpenStax)

Physics reference for the equation, exponential response and time constant.

Equations 14.23–14.25 and 14.27 establish the standard relations. Our identification story, numerical values, examples, graphs and step-by-step prose are newly constructed, not copied figures or exercises from this chapter.

Open the source directly

T3 — Reference and reading scope

Schoukens and Ljung — Nonlinear System Identification: A User-Oriented Roadmap (2019 preprint)

Instructor background for the iterative process and experiment design.

Read The Lead Actors, The Data, Experiment Design and Model Validation. The paper supports matching the data to the task and revisiting earlier choices; it is not an assignment to read the full nonlinear-identification literature before Lecture 1.

Open the source directly

T4 — Reference and reading scope

Munther A. Dahleh — Introductory Examples for System Identification, MIT 6.435, Lecture 2 (2005)

Instructor example of the interaction between model structure and input selection.

Pages/slides 24–29 discuss a step input. Slide 28 explains that a transient can be useful with an appropriate parametric structure. Do not copy its statistical derivations into Lecture 1 or interpret it as universal sufficiency of a step.

Open the source directly

T5 — Reference and reading scope

MathWorks — Validating Models After Estimation

Technical reference for distinct estimation and validation data.

Read Data for Model Validation. It supports the separation of fitting and checking, not a claim that our generated validation demonstrates accuracy on physical hardware. Our illustrative voltage range and switch times are author choices.

Open the source directly

T6 — Reference and reading scope

MathWorks — What Is Residual Analysis?

Terminology safeguard and later instructor reading, not the definition used for all errors in this deck.

This page begins with one-step-ahead prediction residuals. Our M23 defines simulation residuals explicitly. Do not transfer whiteness tests or distribution claims to our plot without the appropriate assumptions. No statistical residual test is taught in the main route.

Open the source directly

T7 — Reference and reading scope

Boyd and Vandenberghe — Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares

Open author-hosted book; language of linear/affine models and next-lecture reading.

Section 2.1 is relevant to linear/affine functions; Chapter 13 is relevant to least-squares fitting. Direct PDF: `https://web.stanford.edu/~boyd/vmls/vmls.pdf` . No textbook illustrations are copied into this package.

Open the source directly

T8 — Reference and reading scope

Lennart Ljung — System Identification: Theory for the User, 2nd edition

Long-course reference, documented as the assigned text on the MIT syllabus.

This link is a syllabus and bibliographic pointer, not the full book. The package does not include or purport to reproduce the textbook.

Open the source directly

U1 — Reference and reading scope

Original Lecture 1: Identification theory — Lecture 1. Introduction

User-provided source; author Alexey Vedyakov, ITMO University.

HDU__Ident__L1-2.pdf: pp. 13–14 contain the RL equation; pp. 51–71 contain the identification workflow. The new text preserves the course purpose and example, but explicitly corrects definitions and expands experiment design.

U2 — Reference and reading scope

Original Lecture 2: Approximation and identification

User-provided source; boundary for the following class.

HDU_Identification_Theory_**Lecture_2**_2026.pdf: pp. 1–40, friction, deterministic least squares and parabola; stochastic section begins p. 41. Do not revise this PDF as part of Lecture 1.